

Tutorial Sheet 4 Solutions

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Problem

- (a) Derive the three-point numerical differentiation formulas for approximating $f'(x)$ (midpoint and endpoint formulas) along with their corresponding error terms.
- (b) Compare the resulting formulas in terms of their error terms and the number of function evaluations required for the approximation.

Solution. Part (a): Derivation of Three-Point Formulas

Let x_0 be the point where we want to evaluate the derivative, and let $h > 0$ be the step size. We use Taylor series expansions of $f(x)$ around x_0 .

1. Three-Point Endpoint Formula (Forward Difference):

Consider the nodes $x_0, x_0 + h$, and $x_0 + 2h$. We expand $f(x_0 + h)$ and $f(x_0 + 2h)$ around x_0 :

$$f(x_0 + h) = f(x_0) + hf'(x_0) + \frac{h^2}{2}f''(x_0) + \frac{h^3}{6}f'''(\xi_1) \quad (1)$$

$$f(x_0 + 2h) = f(x_0) + 2hf'(x_0) + 2h^2f''(x_0) + \frac{8h^3}{6}f'''(\xi_2) \quad (2)$$

where $\xi_1 \in (x_0, x_0 + h)$ and $\xi_2 \in (x_0, x_0 + 2h)$. To eliminate the $f''(x_0)$ term, we multiply equation (1) by 4 and subtract equation (2):

$$4f(x_0 + h) - f(x_0 + 2h) = 3f(x_0) + 2hf'(x_0) + \frac{2h^3}{3}f'''(\xi_1) - \frac{4h^3}{3}f'''(\xi_2)$$

Solving for $f'(x_0)$, we get:

$$f'(x_0) = \frac{-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)}{2h} - \frac{h^2}{3} [2f'''(\xi_2) - f'''(\xi_1)]$$

Using the Intermediate Value Theorem for continuous functions, the error terms can be combined into a single term evaluated at some $\xi \in (x_0, x_0 + 2h)$:

$$f'(x_0) = \frac{-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)}{2h} + \frac{h^2}{3}f'''(\xi) \quad (3)$$

This is the three-point endpoint formula. The error term is $\frac{h^2}{3}f'''(\xi)$.

2. Three-Point Midpoint Formula (Central Difference):

Consider the nodes $x_0 - h, x_0$, and $x_0 + h$. We expand $f(x_0 + h)$ and $f(x_0 - h)$ around x_0 :

$$f(x_0 + h) = f(x_0) + hf'(x_0) + \frac{h^2}{2}f''(x_0) + \frac{h^3}{6}f'''(\xi_1) \quad (4)$$

$$f(x_0 - h) = f(x_0) - hf'(x_0) + \frac{h^2}{2}f''(x_0) - \frac{h^3}{6}f'''(\xi_2) \quad (5)$$

where $\xi_1 \in (x_0, x_0 + h)$ and $\xi_2 \in (x_0 - h, x_0)$. Subtracting equation (5) from equation (4) perfectly eliminates both the $f(x_0)$ and $f''(x_0)$ terms:

$$f(x_0 + h) - f(x_0 - h) = 2hf'(x_0) + \frac{h^3}{6} [f'''(\xi_1) + f'''(\xi_2)]$$

Solving for $f'(x_0)$ and combining the error terms via the Intermediate Value Theorem yields:

$$f'(x_0) = \frac{f(x_0 + h) - f(x_0 - h)}{2h} - \frac{h^2}{6} f'''(\xi) \quad (6)$$

where $\xi \in (x_0 - h, x_0 + h)$. This is the three-point midpoint formula. The error term is $-\frac{h^2}{6} f'''(\xi)$.

Part (b): Comparison

- **Error Term:** Both formulas are of order $\mathcal{O}(h^2)$. However, the coefficient of the error term in the midpoint formula ($\frac{1}{6}$) is half the magnitude of the coefficient in the endpoint formula ($\frac{1}{3}$). Consequently, the midpoint formula generally provides a significantly more accurate approximation.
- **Function Evaluations:** To compute the derivative at x_0 , the midpoint formula requires evaluating the function at **2** points ($x_0 + h$ and $x_0 - h$). The endpoint formula requires evaluating the function at **3** points (x_0 , $x_0 + h$, and $x_0 + 2h$).
- **Conclusion:** The midpoint formula is superior because it yields a smaller theoretical truncation error and requires fewer function evaluations. The endpoint formula is typically reserved for boundaries (the first or last point in a given dataset) where centered points are not available.

□

Problem

The following data are taken from the function $f(x) = x \ln x$.

x	$f(x)$	$f'(x)$
8.1	16.94410	
8.3	17.56492	
8.5	18.19056	
8.7	18.82091	

- Use the most accurate three-point numerical differentiation formulas to approximate $f'(x)$ at each listed value of x .
- Determine appropriate error bounds using the error formulas for the three-point methods, and then compute the actual errors in the approximations obtained in part (a).

Solution. The given function is $f(x) = x \ln x$. The uniform step size is $h = 0.2$. The most accurate 3-point formulas rely on the midpoint (central difference) formula for interior points, and the endpoint (forward/backward difference) formula for the boundaries.

Part (a): Approximating $f'(x)$ **1. For $x = 8.1$ (Left Boundary):**

We use the 3-point forward endpoint formula:

$$f'(x_0) \approx \frac{-3f(x_0) + 4f(x_1) - f(x_2)}{2h}$$

$$f'(8.1) \approx \frac{-3(16.94410) + 4(17.56492) - 18.19056}{2(0.2)}$$

$$\approx \frac{-50.83230 + 70.25968 - 18.19056}{0.4} = \frac{1.23682}{0.4} = \mathbf{3.09205}$$

2. For $x = 8.3$ (Interior):

We use the 3-point midpoint formula:

$$f'(x_1) \approx \frac{f(x_2) - f(x_0)}{2h}$$

$$f'(8.3) \approx \frac{18.19056 - 16.94410}{2(0.2)} = \frac{1.24646}{0.4} = \mathbf{3.11615}$$

3. For $x = 8.5$ (Interior):

We use the 3-point midpoint formula:

$$f'(8.5) \approx \frac{f(8.7) - f(8.3)}{2(0.2)} = \frac{18.82091 - 17.56492}{0.4} = \frac{1.25599}{0.4} = \mathbf{3.139975}$$

4. For $x = 8.7$ (Right Boundary):

We use the 3-point backward endpoint formula:

$$f'(x_n) \approx \frac{3f(x_n) - 4f(x_{n-1}) + f(x_{n-2})}{2h}$$

$$\begin{aligned}
 f'(8.7) &\approx \frac{3(18.82091) - 4(18.19056) + 17.56492}{2(0.2)} \\
 &\approx \frac{56.46273 - 72.76224 + 17.56492}{0.4} = \frac{1.26541}{0.4} = \mathbf{3.163525}
 \end{aligned}$$

Part (b): Error Bounds and Actual Errors

To find the theoretical bounds, we need the third derivative of $f(x) = x \ln x$.

$$\begin{aligned}
 f'(x) &= \ln x + 1 \\
 f''(x) &= \frac{1}{x} \\
 f'''(x) &= -\frac{1}{x^2}
 \end{aligned}$$

Since $|f'''(x)| = \frac{1}{x^2}$ is a strictly decreasing function for $x > 0$, its maximum absolute value on any interval $[a, b]$ will occur at the left endpoint, $x = a$.

Error Bounds:

- **For $x = 8.1$ (Endpoint):** The interval for ξ is $[8.1, 8.5]$.

$$\text{Bound} = \frac{h^2}{3} \max_{\xi \in [8.1, 8.5]} \left| -\frac{1}{\xi^2} \right| = \frac{0.04}{3} \left(\frac{1}{8.1^2} \right) \approx \mathbf{2.032 \times 10^{-4}}$$

- **For $x = 8.3$ (Midpoint):** The interval for ξ is $[8.1, 8.5]$.

$$\text{Bound} = \frac{h^2}{6} \max_{\xi \in [8.1, 8.5]} \left| -\frac{1}{\xi^2} \right| = \frac{0.04}{6} \left(\frac{1}{8.1^2} \right) \approx \mathbf{1.016 \times 10^{-4}}$$

- **For $x = 8.5$ (Midpoint):** The interval for ξ is $[8.3, 8.7]$.

$$\text{Bound} = \frac{h^2}{6} \max_{\xi \in [8.3, 8.7]} \left| -\frac{1}{\xi^2} \right| = \frac{0.04}{6} \left(\frac{1}{8.3^2} \right) \approx \mathbf{9.677 \times 10^{-5}}$$

- **For $x = 8.7$ (Endpoint):** The interval for ξ is $[8.3, 8.7]$.

$$\text{Bound} = \frac{h^2}{3} \max_{\xi \in [8.3, 8.7]} \left| -\frac{1}{\xi^2} \right| = \frac{0.04}{3} \left(\frac{1}{8.3^2} \right) \approx \mathbf{1.935 \times 10^{-4}}$$

Actual Errors:

The true values are computed using $f'(x) = \ln(x) + 1$:

- $f'(8.1) = \ln(8.1) + 1 \approx 3.091864$
- $f'(8.3) = \ln(8.3) + 1 \approx 3.116256$
- $f'(8.5) = \ln(8.5) + 1 \approx 3.140066$
- $f'(8.7) = \ln(8.7) + 1 \approx 3.163323$

We calculate the absolute actual error as $E = |\text{True} - \text{Approximation}|$:

- **At $x = 8.1$:** $|3.091864 - 3.092050| = \mathbf{1.86 \times 10^{-4}}$ (Complies with bound $\leq 2.032 \times 10^{-4}$)
- **At $x = 8.3$:** $|3.116256 - 3.116150| = \mathbf{1.06 \times 10^{-4}}$ (*Slightly exceeds bound $\leq 1.016 \times 10^{-4}$*)
- **At $x = 8.5$:** $|3.140066 - 3.139975| = \mathbf{0.91 \times 10^{-4}}$ (Complies with bound $\leq 0.967 \times 10^{-4}$)
- **At $x = 8.7$:** $|3.163323 - 3.163525| = \mathbf{2.02 \times 10^{-4}}$ (*Slightly exceeds bound $\leq 1.935 \times 10^{-4}$*)

Note: The slight discrepancies where the actual error exceeds the theoretical bound for $x = 8.3$ and $x = 8.7$ are due to the accumulated round-off errors present in the provided five-decimal-place data table, as the theoretical bounds only account for pure truncation error. $\square$

Problem

The following data are taken from the function $f(x) = e^{x/3} + x^2$.

x	$f(x)$	$f'(x)$
-3.0	9.367879	
-2.8	8.233241	
-2.6	7.180350	
-2.4	6.209329	
-2.2	5.320305	
-2.0	4.513417	

- Use the most accurate five-point numerical differentiation formulas to approximate $f'(x)$ at each listed value of x .
- Determine appropriate error bounds using the error formulas for the five-point methods, and then compute the actual errors in the approximations obtained in part (a).

Solution. The given function is $f(x) = e^{x/3} + x^2$. The step size is $h = 0.2$. We are given six data points $x_0, x_1, x_2, x_3, x_4, x_5$.

Part (a): Approximating $f'(x)$

We use the 5-point midpoint formula where centered data is available, and the 5-point endpoint formulas for the boundaries.

• Midpoint Formula (for x_2, x_3):

$$f'(x) \approx \frac{1}{12h} [f(x-2h) - 8f(x-h) + 8f(x+h) - f(x+2h)]$$

• Endpoint Forward Formula (for x_0, x_1):

$$f'(x) \approx \frac{1}{12h} [-25f(x) + 48f(x+h) - 36f(x+2h) + 16f(x+3h) - 3f(x+4h)]$$

• Endpoint Backward Formula (for x_4, x_5):

$$f'(x) \approx \frac{1}{12h} [3f(x-4h) - 16f(x-3h) + 36f(x-2h) - 48f(x-h) + 25f(x)]$$

Applying these formulas ($12h = 2.4$):

• For $x = -3.0$ (Forward using x_0 to x_4):

$$\begin{aligned} f'(-3.0) &\approx \frac{1}{2.4} [-25(9.367879) + 48(8.233241) - 36(7.180350) + 16(6.209329) - 3(5.320305)] \\ &\approx -5.877354 \end{aligned}$$

• For $x = -2.8$ (Forward using x_1 to x_5):

$$\begin{aligned} f'(-2.8) &\approx \frac{1}{2.4} [-25(8.233241) + 48(7.180350) - 36(6.209329) + 16(5.320305) - 3(4.513417)] \\ &\approx -5.468962 \end{aligned}$$

- For $x = -2.6$ (Midpoint using x_0 to x_4):

$$f'(-2.6) \approx \frac{1}{2.4} [9.367879 - 8(8.233241) + 8(6.209329) - 5.320305] \approx -\mathbf{5.059871}$$

- For $x = -2.4$ (Midpoint using x_1 to x_5):

$$f'(-2.4) \approx \frac{1}{2.4} [8.233241 - 8(7.180350) + 8(5.320305) - 4.513417] \approx -\mathbf{4.650381}$$

- For $x = -2.2$ (Backward using x_0 to x_4):

$$f'(-2.2) \approx \frac{1}{2.4} [3(9.367879) - 16(8.233241) + 36(7.180350) - 48(6.209329) + 25(5.320305)] \\ \approx -\mathbf{4.240562}$$

- For $x = -2.0$ (Backward using x_1 to x_5):

$$f'(-2.0) \approx \frac{1}{2.4} [3(8.233241) - 16(7.180350) + 36(6.209329) - 48(5.320305) + 25(4.513417)] \\ \approx -\mathbf{3.828859}$$

Part (b): Error Bounds and Actual Errors

First, we need the fifth derivative of $f(x)$:

$$\begin{aligned} f'(x) &= \frac{1}{3}e^{x/3} + 2x \\ f''(x) &= \frac{1}{9}e^{x/3} + 2 \\ f'''(x) &= \frac{1}{27}e^{x/3} \\ f^{(4)}(x) &= \frac{1}{81}e^{x/3} \\ f^{(5)}(x) &= \frac{1}{243}e^{x/3} \end{aligned}$$

Since $e^{x/3}$ is an increasing function, its maximum absolute value on the interval $[-3.0, -2.0]$ occurs at the rightmost endpoint, $x = -2.0$.

$$\max_{\xi \in [-3.0, -2.0]} |f^{(5)}(\xi)| = \frac{1}{243}e^{-2.0/3} \approx \frac{1}{243}(0.513417) \approx 0.0021128$$

Theoretical Error Bounds:

- **Endpoint Formula Bound** (for $x = -3.0, -2.8, -2.2, -2.0$):

$$\frac{h^4}{5} |f^{(5)}(\xi)| \leq \frac{(0.2)^4}{5} (0.0021128) = \frac{0.0016}{5} (0.0021128) \approx \mathbf{6.76 \times 10^{-7}}$$

- **Midpoint Formula Bound** (for $x = -2.6, -2.4$):

$$\frac{h^4}{30} |f^{(5)}(\xi)| \leq \frac{(0.2)^4}{30} (0.0021128) = \frac{0.0016}{30} (0.0021128) \approx \mathbf{1.13 \times 10^{-7}}$$

Actual Errors: We evaluate the exact derivative $f'(x) = \frac{1}{3}e^{x/3} + 2x$:

- $f'(-3.0) = \frac{1}{3}e^{-1.0} - 6 \approx -5.877373$
- $f'(-2.8) = \frac{1}{3}e^{-2.8/3} - 5.6 \approx -5.468965$
- $f'(-2.6) = \frac{1}{3}e^{-2.6/3} - 5.2 \approx -5.059871$
- $f'(-2.4) = \frac{1}{3}e^{-0.8} - 4.8 \approx -4.650223$
- $f'(-2.2) = \frac{1}{3}e^{-2.2/3} - 4.4 \approx -4.239850$
- $f'(-2.0) = \frac{1}{3}e^{-2.0/3} - 4.0 \approx -3.828861$

Computing the absolute actual errors $|\text{True} - \text{Approximation}|$:

- **At $x = -3.0$:** $|-5.877373 - (-5.877354)| = 1.9 \times 10^{-5}$
- **At $x = -2.8$:** $|-5.468965 - (-5.468962)| = 3.0 \times 10^{-6}$
- **At $x = -2.6$:** $|-5.059871 - (-5.059871)| = 0.0 \times 10^{-6}$
- **At $x = -2.4$:** $|-4.650223 - (-4.650381)| = 1.58 \times 10^{-4}$
- **At $x = -2.2$:** $|-4.239850 - (-4.240562)| = 7.12 \times 10^{-4}$
- **At $x = -2.0$:** $|-3.828861 - (-3.828859)| = 2.0 \times 10^{-6}$

Note: As before, actual errors can exceed truncation theoretical bounds due to the round-off error present in the provided 6-decimal point starting data. $\square$

Problem

Consider the three-point midpoint formula for approximating the derivative

$$f'(x_0) = \frac{f(x_0 + h) - f(x_0 - h)}{2h} - \frac{h^2}{6} f^{(3)}(\xi_1).$$

Suppose that in evaluating $f(x_0 + h)$ and $f(x_0 - h)$ we encounter round-off errors $e(x_0 + h)$ and $e(x_0 - h)$. The computed values $\bar{f}(x_0 + h)$ and $\bar{f}(x_0 - h)$ are related to the true values by

$$\begin{aligned} f(x_0 + h) &= \bar{f}(x_0 + h) + e(x_0 + h), \\ f(x_0 - h) &= \bar{f}(x_0 - h) + e(x_0 - h). \end{aligned}$$

- (a) Derive an error bound for the approximation of $f'(x_0)$ obtained from the three-point midpoint formula when both truncation error and round-off errors are present. Assume $|e(x_0 + h)| \leq \varepsilon$, $|e(x_0 - h)| \leq \varepsilon$, and $|f^{(3)}(x)| \leq M$ for x in the relevant interval.
- (b) Based on the derived bound, discuss whether it is practical to take h as small as possible. Determine an appropriate choice of h that balances truncation and round-off errors.

Solution. Part (a): Deriving the Error Bound

Let the approximation of the derivative using the computed (numerical) values be denoted by $\tilde{f}'(x_0)$:

$$\tilde{f}'(x_0) = \frac{\bar{f}(x_0 + h) - \bar{f}(x_0 - h)}{2h}$$

The exact mathematical relationship, taking into account the truncation error, is:

$$f'(x_0) = \frac{f(x_0 + h) - f(x_0 - h)}{2h} - \frac{h^2}{6} f^{(3)}(\xi_1)$$

Substituting the expressions for the true values $f(x_0 + h)$ and $f(x_0 - h)$ in terms of their computed values and round-off errors:

$$\begin{aligned} f'(x_0) &= \frac{[\bar{f}(x_0 + h) + e(x_0 + h)] - [\bar{f}(x_0 - h) + e(x_0 - h)]}{2h} - \frac{h^2}{6} f^{(3)}(\xi_1) \\ &= \frac{\bar{f}(x_0 + h) - \bar{f}(x_0 - h)}{2h} + \frac{e(x_0 + h) - e(x_0 - h)}{2h} - \frac{h^2}{6} f^{(3)}(\xi_1) \\ &= \tilde{f}'(x_0) + \frac{e(x_0 + h) - e(x_0 - h)}{2h} - \frac{h^2}{6} f^{(3)}(\xi_1) \end{aligned}$$

The total error is the difference between the true derivative and the computed approximation:

$$E(h) = f'(x_0) - \tilde{f}'(x_0) = \frac{e(x_0 + h) - e(x_0 - h)}{2h} - \frac{h^2}{6} f^{(3)}(\xi_1)$$

Taking the absolute value and applying the triangle inequality:

$$|E(h)| \leq \frac{|e(x_0 + h)| + |e(x_0 - h)|}{2h} + \frac{h^2}{6} |f^{(3)}(\xi_1)|$$

Given that $|e(x_0 + h)| \leq \varepsilon$, $|e(x_0 - h)| \leq \varepsilon$, and $|f^{(3)}(\xi_1)| \leq M$, we obtain the total error bound:

$$|E(h)| \leq \frac{\varepsilon + \varepsilon}{2h} + \frac{h^2}{6} M = \frac{\varepsilon}{h} + \frac{Mh^2}{6} \quad (7)$$

Part (b): Optimal Choice of h

Let the upper bound of the total error be denoted by $B(h)$:

$$B(h) = \frac{\varepsilon}{h} + \frac{M}{6}h^2$$

Is it practical to take h as small as possible?

No, it is not practical. As $h \rightarrow 0$, the truncation error portion $(\frac{M}{6}h^2)$ approaches zero, but the round-off error portion $(\frac{\varepsilon}{h})$ approaches infinity. Thus, making h arbitrarily small will cause the total error to blow up because dividing the inherent machine precision limits (ε) by a tiny number amplifies the error.

Determining the appropriate choice of h :

To find the step size h that minimizes the maximum error bound $B(h)$, we take the derivative of $B(h)$ with respect to h and set it to zero:

$$B'(h) = -\frac{\varepsilon}{h^2} + \frac{M}{3}h = 0$$

Solving for h :

$$\begin{aligned}\frac{M}{3}h &= \frac{\varepsilon}{h^2} \\ h^3 &= \frac{3\varepsilon}{M} \\ h &= \left(\frac{3\varepsilon}{M}\right)^{1/3}\end{aligned}$$

To verify this is a minimum, we check the second derivative:

$$B''(h) = \frac{2\varepsilon}{h^3} + \frac{M}{3}$$

Since $\varepsilon, M, h > 0$, it follows that $B''(h) > 0$, confirming that $h = \left(\frac{3\varepsilon}{M}\right)^{1/3}$ yields the minimum error bound. This is the optimal step size h that perfectly balances the truncation error against the machine round-off error. $\square$

Problem

Use Heun's method of order two to approximate the solution of the initial-value problem

$$y' = te^{3t} - 2y, \quad 0 \leq t \leq 1, \quad y(0) = 0$$

with step size $h = 0.5$.

Solution. The initial-value problem is defined by $f(t, y) = te^{3t} - 2y$, with initial conditions $t_0 = 0$, $w_0 = y(0) = 0$, and a step size of $h = 0.5$. We need to find approximations w_1 at $t_1 = 0.5$ and w_2 at $t_2 = 1.0$.

Heun's method of order two (also known as the explicit trapezoidal rule) consists of a predictor step (Euler's method) and a corrector step:

$$\text{Predictor: } \tilde{w}_{i+1} = w_i + hf(t_i, w_i)$$

$$\text{Corrector: } w_{i+1} = w_i + \frac{h}{2} [f(t_i, w_i) + f(t_{i+1}, \tilde{w}_{i+1})]$$

Step 1: Approximation at $t_1 = 0.5$

- Evaluate $f(t_0, w_0)$:

$$f(0, 0) = 0 \cdot e^{3(0)} - 2(0) = 0$$

- Compute the predictor $\tilde{w}_1$:

$$\tilde{w}_1 = w_0 + hf(0, 0) = 0 + 0.5(0) = 0$$

- Evaluate $f(t_1, \tilde{w}_1)$ at $t_1 = 0.5$:

$$f(0.5, 0) = 0.5e^{3(0.5)} - 2(0) = 0.5e^{1.5} \approx 2.240845$$

- Compute the corrector w_1 :

$$w_1 = w_0 + \frac{0.5}{2} [f(0, 0) + f(0.5, 0)]$$

$$w_1 \approx 0 + 0.25[0 + 2.240845] = \mathbf{0.560211}$$

Thus, the approximation at $t = 0.5$ is $y(0.5) \approx 0.560211$.

Step 2: Approximation at $t_2 = 1.0$

- Evaluate $f(t_1, w_1)$:

$$f(0.5, 0.560211) = 0.5e^{1.5} - 2(0.560211) \approx 2.240845 - 1.120422 = 1.120423$$

- Compute the predictor $\tilde{w}_2$:

$$\tilde{w}_2 = w_1 + hf(t_1, w_1) \approx 0.560211 + 0.5(1.120423) = 1.120423$$

- Evaluate $f(t_2, \tilde{w}_2)$ at $t_2 = 1.0$:

$$f(1.0, 1.120423) = 1.0e^{3(1.0)} - 2(1.120423) \approx 20.085537 - 2.240846 = 17.844691$$

- Compute the corrector w_2 :

$$w_2 = w_1 + \frac{0.5}{2}[f(t_1, w_1) + f(t_2, \tilde{w}_2)]$$

$$w_2 \approx 0.560211 + 0.25[1.120423 + 17.844691]$$

$$w_2 \approx 0.560211 + 0.25[18.965114] \approx 0.560211 + 4.741279 = \mathbf{5.301490}$$

Final Answer:

The approximate solutions are $w_1 \approx \mathbf{0.560211}$ at $t = 0.5$, and $w_2 \approx \mathbf{5.301490}$ at $t = 1.0$. $\square$

Problem

Suppose f is continuous and satisfies a Lipschitz condition with constant L on $D = \{(t, y) \mid a \leq t \leq b, -\infty < y < \infty\}$, and suppose there exists a constant M such that $|y''(t)| \leq M$ for all $t \in [a, b]$, where $y(t)$ is the unique solution of the initial-value problem

$$y' = f(t, y), \quad a \leq t \leq b, \quad y(a) = \alpha$$

- (a) Let $w_0, w_1, \dots, w_N$ be the approximations generated by Euler's method with step size h . Prove that for each $i = 0, 1, 2, \dots, N$,

$$|y(t_i) - w_i| \leq \frac{hM}{2L}(e^{L(t_i-a)} - 1).$$

- (b) Suppose the approximations $u_0, u_1, \dots, u_N$ are generated by the perturbed Euler method

$$u_0 = \alpha + \delta_0, \quad u_{i+1} = u_i + hf(t_i, u_i) + \delta_{i+1}, \quad i = 0, 1, \dots, N-1$$

where δ_i represents the round-off error at each step and $|\delta_i| \leq \delta$. Show that for each $i = 0, 1, \dots, N$;

$$|y(t_i) - u_i| \leq \frac{1}{L} \left(\frac{hM}{2} + \frac{\delta}{h} \right) [e^{L(t_i-a)} - 1] + |\delta_0|e^{L(t_i-a)}.$$

Using this bound, discuss the effect of round-off errors and determine an appropriate choice of the step size h .

Solution. To prove both parts, we will use the following well-known lemma for difference inequalities:

Lemma: If $s > 0$ and $t > 0$, and a sequence $\{a_i\}_{i=0}^N$ satisfies $a_{i+1} \leq (1+s)a_i + t$, then

$$a_i \leq e^{is}a_0 + \frac{t}{s}(e^{is} - 1) \quad (8)$$

Also note that $is = i(hL) = (ih)L = L(t_i - a)$, since $t_i = a + ih$.

Part (a): Error Bound for Standard Euler's Method

Using Taylor's Theorem, the exact solution $y(t)$ satisfies:

$$y(t_{i+1}) = y(t_i) + hf(t_i, y(t_i)) + \frac{h^2}{2}y''(\xi_i) \quad (9)$$

for some $\xi_i \in (t_i, t_{i+1})$. Euler's method generates approximations via:

$$w_{i+1} = w_i + hf(t_i, w_i) \quad (10)$$

Subtracting (10) from (9) gives the error equation:

$$y(t_{i+1}) - w_{i+1} = y(t_i) - w_i + h[f(t_i, y(t_i)) - f(t_i, w_i)] + \frac{h^2}{2}y''(\xi_i)$$

Taking the absolute value and applying the triangle inequality:

$$|y(t_{i+1}) - w_{i+1}| \leq |y(t_i) - w_i| + h|f(t_i, y(t_i)) - f(t_i, w_i)| + \frac{h^2}{2}|y''(\xi_i)|$$

Since f satisfies a Lipschitz condition with constant L , $|f(t_i, y(t_i)) - f(t_i, w_i)| \leq L|y(t_i) - w_i|$. We are also given $|y''(t)| \leq M$. Let $e_i = |y(t_i) - w_i|$. Substituting these bounds yields:

$$e_{i+1} \leq e_i + hLe_i + \frac{h^2M}{2} = (1 + hL)e_i + \frac{h^2M}{2}$$

Now we apply the Lemma (8) with $a_i = e_i$, $s = hL$, and $t = \frac{h^2M}{2}$. Since the initial condition is exact in the standard Euler method, $w_0 = \alpha = y(t_0)$, meaning $e_0 = 0$.

$$e_i \leq e^{i(hL)}(0) + \frac{h^2M/2}{hL}(e^{i(hL)} - 1)$$

$$|y(t_i) - w_i| \leq \frac{hM}{2L}(e^{L(t_i-a)} - 1)$$

This completes the proof for part (a).

Part (b): Error Bound for Perturbed Euler's Method

The perturbed Euler method is given by:

$$u_{i+1} = u_i + hf(t_i, u_i) + \delta_{i+1} \quad (11)$$

Subtracting (11) from the exact Taylor expansion (9):

$$y(t_{i+1}) - u_{i+1} = y(t_i) - u_i + h[f(t_i, y(t_i)) - f(t_i, u_i)] + \frac{h^2}{2}y''(\xi_i) - \delta_{i+1}$$

Taking the absolute value, applying the triangle inequality, and using the Lipschitz condition and bounds $|y''(t)| \leq M$ and $|\delta_i| \leq \delta$:

$$|y(t_{i+1}) - u_{i+1}| \leq |y(t_i) - u_i| + hL|y(t_i) - u_i| + \frac{h^2}{2}M + \delta$$

$$|y(t_{i+1}) - u_{i+1}| \leq (1 + hL)|y(t_i) - u_i| + \left(\frac{h^2M}{2} + \delta\right)$$

Let $E_i = |y(t_i) - u_i|$. Applying the Lemma (8) with $a_i = E_i$, $s = hL$, and $t = \frac{h^2M}{2} + \delta$:

$$E_i \leq e^{i(hL)}E_0 + \frac{\frac{h^2M}{2} + \delta}{hL}(e^{i(hL)} - 1)$$

Here, the initial error is $E_0 = |y(t_0) - u_0| = |\alpha - (\alpha + \delta_0)| = |\delta_0|$. Replacing $i(hL)$ with $L(t_i - a)$:

$$|y(t_i) - u_i| \leq e^{L(t_i-a)}|\delta_0| + \frac{1}{L} \left(\frac{hM}{2} + \frac{\delta}{h} \right) [e^{L(t_i-a)} - 1]$$

This completes the proof for the perturbed error bound.

Discussion on Round-off Errors and Step Size:

Let the term dependent on h in the error bound be denoted by $E(h)$:

$$E(h) = \frac{hM}{2} + \frac{\delta}{h}$$

- **Effect of round-off errors:** As the step size h decreases ($h \rightarrow 0$), the mathematical truncation error $\frac{hM}{2}$ approaches zero. However, the accumulated round-off error term $\frac{\delta}{h}$ grows without bound (approaching infinity). Therefore, unlike the idealized mathematical method, we cannot make the actual error arbitrarily small simply by shrinking h .

- **Appropriate choice of step size h :** To minimize the maximum error bound $E(h)$, we find its critical point by setting its derivative with respect to h to zero:

$$\begin{aligned}E'(h) &= \frac{M}{2} - \frac{\delta}{h^2} = 0 \\ \frac{\delta}{h^2} &= \frac{M}{2} \\ h^2 &= \frac{2\delta}{M} \implies h = \sqrt{\frac{2\delta}{M}}\end{aligned}$$

Since $E''(h) = \frac{2\delta}{h^3} > 0$ for positive h , this critical point represents a minimum. Thus, the optimal step size that balances truncation error and round-off error is $h = \sqrt{\frac{2\delta}{M}}$.

□

Problem

Use the Modified Euler method and the Runge-Kutta method of order four to approximate the solution of the initial-value problem

$$y' = te^{3t} - 2y, \quad 0 \leq t \leq 1, \quad y(0) = 0$$

with step size $h = 0.5$. Compare the numerical results with the exact solution

$$y(t) = \frac{1}{5}te^{3t} - \frac{1}{25}e^{3t} + \frac{1}{25}e^{-2t}.$$

Solution. The initial-value problem is defined by $f(t, y) = te^{3t} - 2y$, with $t_0 = 0, w_0 = 0$, and step size $h = 0.5$. We need approximations at $t_1 = 0.5$ and $t_2 = 1.0$.

1. Modified Euler Method

The Modified Euler method (also known as the Midpoint Method) is given by:

$$w_{i+1} = w_i + hf\left(t_i + \frac{h}{2}, w_i + \frac{h}{2}f(t_i, w_i)\right)$$

Step 1: Approximation at $t_1 = 0.5$

$$\begin{aligned} f(t_0, w_0) &= f(0, 0) = 0 \\ w_1 &= 0 + 0.5f(0 + 0.25, 0 + 0.25(0)) \\ &= 0.5f(0.25, 0) = 0.5 \left[0.25e^{3(0.25)} - 2(0) \right] \\ &= 0.5[0.25e^{0.75}] \approx 0.5[0.529250] = \mathbf{0.264625} \end{aligned}$$

Step 2: Approximation at $t_2 = 1.0$

$$\begin{aligned} f(t_1, w_1) &= f(0.5, 0.264625) = 0.5e^{1.5} - 2(0.264625) \approx 2.240845 - 0.529250 = 1.711595 \\ w_2 &= 0.264625 + 0.5f(0.5 + 0.25, 0.264625 + 0.25(1.711595)) \\ &= 0.264625 + 0.5f(0.75, 0.692524) \\ &= 0.264625 + 0.5 \left[0.75e^{3(0.75)} - 2(0.692524) \right] \\ &\approx 0.264625 + 0.5 [7.115801 - 1.385048] \approx 0.264625 + 0.5(5.730753) = \mathbf{3.130001} \end{aligned}$$

2. Runge-Kutta Method of Order Four (RK4)

The RK4 method is given by $w_{i+1} = w_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$, where:

$$\begin{aligned} k_1 &= hf(t_i, w_i) \\ k_2 &= hf(t_i + h/2, w_i + k_1/2) \\ k_3 &= hf(t_i + h/2, w_i + k_2/2) \\ k_4 &= hf(t_{i+1}, w_i + k_3) \end{aligned}$$

Step 1: Approximation at $t_1 = 0.5$

$$\begin{aligned} k_1 &= 0.5f(0, 0) = 0 \\ k_2 &= 0.5f(0.25, 0) = 0.5 [0.25e^{0.75}] \approx 0.264625 \\ k_3 &= 0.5f(0.25, 0.132313) = 0.5 [0.25e^{0.75} - 2(0.132313)] \approx 0.132312 \\ k_4 &= 0.5f(0.5, 0.132312) = 0.5 [0.5e^{1.5} - 2(0.132312)] \approx 0.988110 \\ w_1 &= 0 + \frac{1}{6}(0 + 2(0.264625) + 2(0.132312) + 0.988110) \approx \mathbf{0.296997} \end{aligned}$$

Step 2: Approximation at $t_2 = 1.0$

$$k_1 = 0.5f(0.5, 0.296997) \approx 0.5[2.240845 - 0.593994] \approx 0.823425$$

$$k_2 = 0.5f(0.75, 0.708710) \approx 0.5[7.115801 - 1.417420] \approx 2.849190$$

$$k_3 = 0.5f(0.75, 1.721592) \approx 0.5[7.115801 - 3.443184] \approx 1.836308$$

$$k_4 = 0.5f(1.0, 2.133305) \approx 0.5[20.085537 - 4.266610] \approx 7.909464$$

$$w_2 = 0.296997 + \frac{1}{6}(0.823425 + 2(2.849190) + 2(1.836308) + 7.909464) \approx \mathbf{3.314311}$$

3. Exact Solution and Comparison

We evaluate the exact solution $y(t) = \frac{1}{5}te^{3t} - \frac{1}{25}e^{3t} + \frac{1}{25}e^{-2t}$ at $t = 0.5$ and $t = 1.0$:

$$y(0.5) = 0.1e^{1.5} - 0.04e^{1.5} + 0.04e^{-1} = 0.06e^{1.5} + 0.04e^{-1} \approx \mathbf{0.283617}$$

$$y(1.0) = 0.2e^3 - 0.04e^3 + 0.04e^{-2} = 0.16e^3 + 0.04e^{-2} \approx \mathbf{3.219099}$$

Summary Comparison Table:

t	Exact $y(t)$	Mod. Euler w_i	ME Error	RK4 w_i	RK4 Error
0.5	0.283617	0.264625	0.018992	0.296997	0.013380
1.0	3.219099	3.130001	0.089098	3.314311	0.095212

Note: The relatively large step size $h = 0.5$ leads to significant truncation errors in both methods over the interval, particularly since the function contains rapidly growing exponential terms (e^{3t}). $\square$

Problem

Consider the difference method

$$w_0 = \alpha,$$

$$w_{i+1} = w_i + a_1 f(t_i, w_i) + a_2 f(t_i + \alpha_2, w_i + \delta_2 f(t_i, w_i)), \quad \text{for } i = 0, 1, \dots, N-1$$

Show that this method cannot have local truncation error $O(h^3)$ for any choice of the constants a_1, a_2, α_2 , and δ_2 .

Solution. The local truncation error $\tau_{i+1}(h)$ is defined by replacing the approximation w_i with the exact solution $y(t_i)$ and dividing the residual by h :

$$\tau_{i+1}(h) = \frac{y(t_{i+1}) - y(t_i)}{h} - \frac{1}{h} \left[a_1 f(t_i, y(t_i)) + a_2 f(t_i + \alpha_2, y(t_i) + \delta_2 f(t_i, y(t_i))) \right]$$

For the method to have a local truncation error of $O(h^3)$, the numerator must match the exact solution expansion up to the $O(h^3)$ terms. That is, we must have:

$$y(t_{i+1}) - y(t_i) - \left[a_1 f + a_2 f(t_i + \alpha_2, y_i + \delta_2 f) \right] = O(h^4)$$

where f and its derivatives are evaluated at $(t_i, y(t_i))$.

1. Taylor Expansion of the Exact Solution

We compute the exact increment $y(t_{i+1}) - y(t_i)$ using the Taylor series of $y(t)$ around t_i . Since $y'(t) = f(t, y(t))$, we can find the higher derivatives using the chain rule:

$$\begin{aligned} y' &= f \\ y'' &= f_t + f_y y' = f_t + f f_y \\ y''' &= \frac{d}{dt}(f_t + f f_y) = (f_{tt} + f_{ty} f) + f(f_{yt} + f_{yy} f) + f_y(f_t + f_y f) \\ &= f_{tt} + 2f f_{ty} + f^2 f_{yy} + f_t f_y + f f_y^2 \end{aligned}$$

Substitute these into the Taylor series $y(t_{i+1}) = y(t_i) + h y' + \frac{h^2}{2} y'' + \frac{h^3}{6} y''' + O(h^4)$:

$$y(t_{i+1}) - y(t_i) = h f + \frac{h^2}{2} (f_t + f f_y) + \frac{h^3}{6} (f_{tt} + 2f f_{ty} + f^2 f_{yy} + f_t f_y + f f_y^2) + O(h^4) \quad (12)$$

2. Taylor Expansion of the Numerical Method

Now we expand the numerical increment $T = a_1 f + a_2 f(t_i + \alpha_2, y(t_i) + \delta_2 f)$ using the multivariable Taylor series for f around $(t_i, y(t_i))$:

$$f(t_i + \alpha_2, y_i + \delta_2 f) = f + \alpha_2 f_t + \delta_2 f f_y + \frac{1}{2} (\alpha_2^2 f_{tt} + 2\alpha_2 \delta_2 f f_{ty} + \delta_2^2 f^2 f_{yy}) + O(\text{order } 3)$$

Multiplying by a_2 and adding $a_1 f$, the numerical increment is:

$$T = (a_1 + a_2) f + a_2 \alpha_2 f_t + a_2 \delta_2 f f_y + \frac{1}{2} a_2 \alpha_2^2 f_{tt} + a_2 \alpha_2 \delta_2 f f_{ty} + \frac{1}{2} a_2 \delta_2^2 f^2 f_{yy} + \dots \quad (13)$$

3. Comparison of Coefficients

For the method to be $O(h^3)$, the expansions in (12) and (13) must be identical for any arbitrary, sufficiently smooth function $f(t, y)$.

Comparing the $O(h^3)$ terms from the exact solution to the numerical method's expansion:

- The exact expansion (12) contains the terms $\frac{h^3}{6}f_tf_y$ and $\frac{h^3}{6}ff_y^2$.
- The numerical expansion (13) evaluates f at a single advanced point. As a result, its Taylor expansion produces purely second partial derivatives (f_{tt}, f_{ty}, f_{yy}) at the quadratic level. It **structurally cannot** generate products of first-order partial derivatives like f_tf_y or ff_y^2 .

Since there are no parameters $a_1, a_2, \alpha_2, \delta_2$ that can generate the required $\frac{h^3}{6}f_y(f_t + ff_y)$ terms to match the exact solution's y''' component, the expansions will always differ by at least $\mathcal{O}(h^3)$.

Therefore, $y(t_{i+1}) - w_{i+1} = \mathcal{O}(h^3)$, which means dividing by h leaves a local truncation error of $\tau_{i+1}(h) = \mathcal{O}(h^2)$. The method cannot have a local truncation error of $\mathcal{O}(h^3)$. $\square$